<u>Practical – 04</u>

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Division: CS1 **Batch:**C14

4.1.1. Pandas - series creation and manipulation

Write a Python program that takes a list of numbers from the user, creates a Pandas series from it, and then calculates the mean of even and odd numbers separately using the groupby and mean() operations.

Hint: You can group the Series based on whether each number is even or odd.

Input Format:

The user should enter a list of numbers separated by space when prompted.

Output Format:

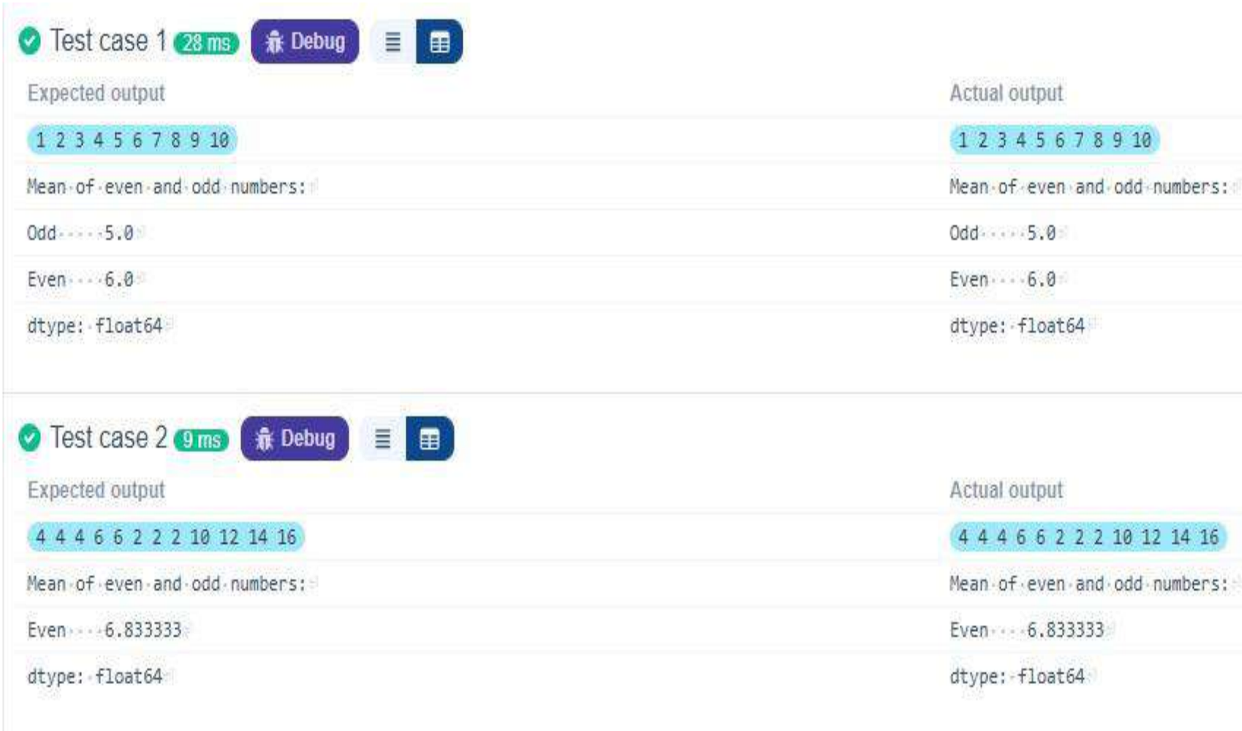
The program should display the mean of even and odd numbers separately.

Each mean value should be displayed with a label indicating whether it corresponds to even or odd numbers.

Code:

```
import pandas as pd

 # Take inputs from the user to create a list of numbers numbers = list(map(int,

input().split()))

 # Create a Pandas series from the list of numbers series=pd.Series(numbers)

 # Grouping by even and odd numbers and calculating the mean

grouped = series.groupby(series%2==0).mean()  # Display the mean of even and odd numbers

with labels grouped.index = ['Even' if is_even else 'Odd' for is_even in grouped.index]

print("Mean of even and odd numbers:") print(grouped)
```

Output:



4.1.2. Dictionary to dataframe

A dictionary of lists has been provided to you in the editor. Create a DataFrame from the dictionary of lists and perform the listed operations, then display the DataFrame before and after each manipulation.

Create the DataFrame:

Convert the dictionary to a Pandas DataFrame.

Add a new row:

Take inputs from the user for the new row data (name, age).

Add the new row to the DataFrame.

Display the DataFrame after adding the new row.

Modify a row:

Modify a specific row by changing the age. Take the row index and new age value from the user.

Display the DataFrame after modifying the row.

Delete a row:

Take the row index to be deleted from the user.

Remove the specified row.

Display the DataFrame after deleting the row.

Add a new column:

Add a column Gender with values taken from the user.

Display the DataFrame after adding the new column.

Modify a column:

Convert names to uppercase.

Display the DataFrame after modifying the column.

Delete a column:

Remove the Age column.

Display the DataFrame after deleting the column.

Code:

```
import pandas as pd

# Provided dictionary of lists data = {

   'Name': ['Alice', 'Bob', 'Charlie'],

   'Age': [25, 30, 35],

}

# Convert the dictionary to a DataFrame df =

pd.DataFrame(data)

# Display the original DataFrame

print("Original DataFrame:") print(df)

name = input("New name: ") age =int(input("New age:

")) newrow= {"Name": name,"Age":age} df
```

```
=df.append(newrow, ignore_index = True) print("After
adding a row:\n",df)
# Modifying a row ind = int(input("Index of row to
modify: ")) age = int(input("New age: "))
df.loc[ind,"Age"] = age
# Display the DataFrame after modifying a row print("After modifying a row:") print(df)
# Deleting a row ind =int(input("Index of row to delete: ")) df =
df.drop(index = ind).reset_index(drop = True) # Display the
DataFrame after deleting a row print("After deleting a row:")
print(df)
# Adding a new column gender = input("Enter genders separated by space:
").split() df["Gender"]=gender print("After adding a new column:") print(df)
# Modifying a column
column = [] for i in
df["Name"]:
column.append(i.upper()) df["Name"] =
column
# Display the DataFrame after modifying a column print("After
modifying a column:") print(df)

# Deleting a column df =
df.drop(columns = ["Age"])
# Display the DataFrame after deleting a column print("After
deleting a column:") print(df)
```

Output:

Test case 1 94 ms Debug

Expected output	Actual output
Original DataFrame:	Original DataFrame:
Name Age	Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
New name: Susan	New name: Susan
New age: 29	New age: 29
After adding a row:	After adding a row:
Name Age	Name Age
0 Alice 25	0 Alice 25
1 Bob 30	1 Bob 30
2 Charlie 35	2 Charlie 35
3 Susan 29	3 Susan 29
Index of row to modify: 0	Index of row to modify: 0

```
New age: 27
After modifying a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to delete: 3
After deleting a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
Enter genders separated by space: F M M
After adding a new column:
      Name  Age Gender
0    Alice   27      F
1      Bob   30      M
2  Charlie   35      M
After modifying a column:
      Name  Age Gender
0    ALICE   27      F
1      BOB   30      M
2  CHARLIE   35      M
After deleting a column:
      Name Gender
0    ALICE      F
1      BOB      M
2  CHARLIE      M
```


```
New age: 27
After modifying a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
3    Susan   29
Index of row to delete: 3
After deleting a row:
      Name  Age
0    Alice   27
1      Bob   30
2  Charlie   35
Enter genders separated by space: F M M
After adding a new column:
      Name  Age Gender
0    Alice   27      F
1      Bob   30      M
2  Charlie   35      M
After modifying a column:
      Name  Age Gender
0    ALICE   27      F
1      BOB   30      M
2  CHARLIE   35      M
After deleting a column:
      Name Gender
0    ALICE      F
1      BOB      M
2  CHARLIE      M
```


4.1.3. Student Information

Write a program to read a text file containing student information (name, age, and grade) using Pandas. Perform the following tasks:

Display the first five rows of the data frame.

Calculate the average age of the students (limit the average age up to 2 decimal places).

Filter out the students who have a grade above a certain threshold (consider the threshold grade is 'B').

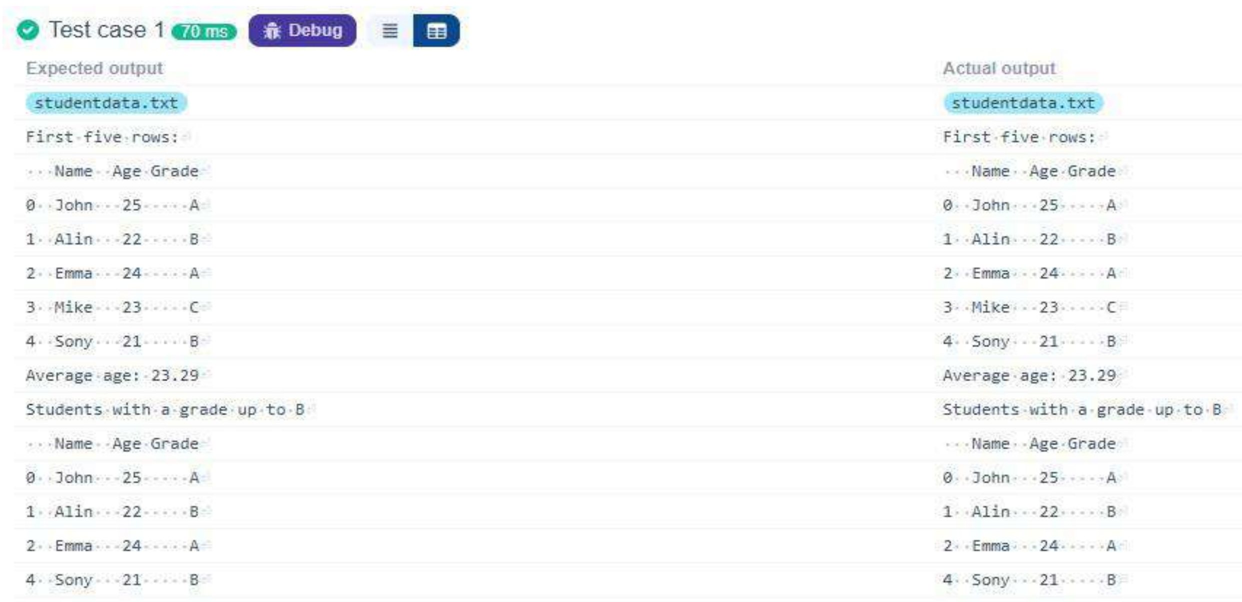
Code:

import pandas as pd

Read the text file into a DataFrame file = input()

```
data = pd.read_csv(file, sep="\s+", header=None, names=["Name", "Age", "Grade"])
 # write your code here..  print("First five rows:")
print(data.head())  avg_age =
round(data["Age"].mean(),2)  print(f"Average age:
{avg_age}")  print("Students with a grade up to B")
filtered_student = data[data["Grade"] <= "B"]
print(filtered_student)
```

Output:



4.2.1. Month with the Highest Total Sales

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by Month and calculate the total sales for each month.

Find the month with the highest total sales and display it.

Also, display the total sales for the best month.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
 # Prompt the user for the file name file_name = input()  # Load the data df =
pd.read_csv(file_name)  df["Date"] = pd.to_datetime(df["Date"])
df["Month"] = df["Date"].dt.strftime("%Y-%m") df["Total Sales"] =
df["Quantity"] * df["Price"]  sales_by_month = df.groupby("Month")["Total
Sales"].sum() best_month = sales_by_month.idxmax() highest_sales =
sales_by_month.max() print(f"Best month: {best_month}") print(f"Total
sales: ${highest_sales:.2f}")
```

Output:

4.2.2. Best Selling Product

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Find the product that sold the most in terms of quantity sold.

Display the product that sold the most and the total quantity sold for that product.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
 # Prompt the user for the file name file_name = input()  # Load the
data df = pd.read_csv(file_name) product_sales =
df.groupby("Product")["Quantity"].sum()  best_product =
product_sales.idxmax() highest_quantity = product_sales.max()
 # Display the result
print(f"Best selling product: {best_product}") print(f"Total quantity
sold: {highest_quantity}")
```

Output:

4.2.3. City that Sold the Most Products

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the columns: Date, Product, Quantity, Price, and City.

Group the data by City and calculate the total quantity of products sold for each city.

Find the city that sold the most products (based on the total quantity sold).

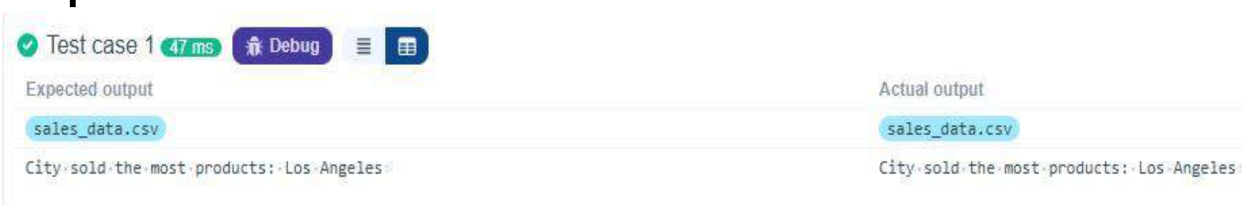
Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago
2025-01-05,Product C,10,30,Los Angeles
```

Code:

```
import pandas as pd
 # Prompt the user for the file name
file_name = input()  # Load the data df =
pd.read_csv(file_name)
 city_sales=df.groupby("City")["Quantity"].sum() best_city=city_sales.idxmax()
# Display the result print(f"City sold the most products:
{best_city}")
```

Output:



4.2.4. Most Frequently Sold Product Pairs

Write a Python program that takes the file name of a CSV file as input, reads the data, and performs the following operations:

The CSV file contains the following columns: Date, Product, Quantity, Price, and City.

For each date, find all pairs of products that were sold together (i.e., two products sold on the same date).

Output the product pair/s that was sold most frequently.

Sample Data:

```
Date,Product,Quantity,Price,City
2025-01-01,Product A,5,20,New York
```

```
2025-01-01,Product B,3,15,Los Angeles
2025-01-02,Product A,7,20,New York
2025-01-02,Product C,4,30,Chicago
2025-01-03,Product B,2,15,Chicago
2025-01-03,Product A,8,20,Los Angeles 2025-01-
04,Product C,6,30,New York
2025-01-04,Product B,5,15,Los Angeles
2025-01-05,Product A,3,20,Chicago 2025-01-05,Product
C,10,30,Los Angeles
```

Code:

```
import pandas as pd from itertools import
combinations from collections import Counter
# Prompt user to input the file name
file_name = input()
 # Read data from the specified CSV file
df = pd.read_csv(file_name)
group=df.groupby("Date")["Product"].apply(list)
product_pair=[] for product_pair in group:
grouped = df.groupby("Date")["Product"].apply(list)
product_pair = []  for products in grouped: if len(products)>1:
product_pair.extend(combinations(sorted(products),2))
pair_counts =Counter(product_pair)  if pair_counts:
max_count = max(pair_counts.values()) most_common_pairs = [pair for pair,count in
pair_counts.items() if count == max_count] for pair in sorted(most_common_pairs):
print(f"{pair[0]} and {pair[1]}: {max_count} times")
```

Output:

4.2.5. Titanic Dataset Analysis and Data Cleaning

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset. For each question, perform necessary data cleaning, transformations, and calculations as required.

Display the first 5 rows of the dataset.

Display the last 5 rows of the dataset.

Get the shape of the dataset (number of rows and columns).

Get a summary of the dataset (using .info()).

Get basic statistics (mean, standard deviation, etc.) of the dataset using .describe().

Check for missing values and display the count of missing values for each column.

Fill missing values in the 'Age' column with the median age.

Fill missing values in the 'Embarked' column with the most frequent value (mode()).

Drop the 'Cabin' column due to many missing values.

Create a new column, 'FamilySize' by adding the 'SibSp' and 'Parch' columns.

The Titanic dataset contains columns as shown below,

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

import pandas as pd import numpy as np # Load the

Titanic dataset data = pd.read_csv('Titanic-

Dataset.csv') # 1. Display the first 5 rows of the

dataset print(data.head())

2. Display the last 5 rows of the dataset print(data.tail())

3. Get the shape of the dataset print(data.shape)

4. Get a summary of the dataset (info) print(data.info())

5. Get basic statistics of the dataset

print(data.describe()) # 6. Check for missing

values print(data.isnull().sum())

7. Fill missing values in the 'Age' column with the median age

data['Age'].fillna(data['Age'].median(),inplace=True)

8. Fill missing values in the 'Embarked' column with the mode

data['Embarked'].fillna(data['Age'].mode()[0],inplace=True) # 9. Drop the 'Cabin'

column due to many missing values data.drop(columns=['Cabin'],inplace=True)

10. Create a new column 'FamilySize' by adding 'SibSp' and 'Parch'

data['Familysize']=data['SibSp']+ data['Parch']

Output:

Test case 1 239 ms Debug

Expected output	Actual output
PassengerId Survived Pclass ... Fare Cabin Embarked	PassengerId Survived Pclass ... Fare Cabin Embarked
0 1 0 3 ... 7.2500 NaN S	0 1 0 3 ... 7.2500 NaN S
1 2 1 1 ... 71.2833 C85 C	1 2 1 1 ... 71.2833 C85 C
2 3 1 3 ... 7.9250 NaN S	2 3 1 3 ... 7.9250 NaN S
3 4 1 1 ... 53.1000 C123 S	3 4 1 1 ... 53.1000 C123 S
4 5 0 3 ... 8.0500 NaN S	4 5 0 3 ... 8.0500 NaN S
[5 rows x 12 columns]	[5 rows x 12 columns]
PassengerId Survived Pclass ... Fare Cabin Embarked	PassengerId Survived Pclass ... Fare Cabin Embarked
886 887 0 2 ... 13.00 NaN S	886 887 0 2 ... 13.00 NaN S
887 888 1 1 ... 30.00 B42 S	887 888 1 1 ... 30.00 B42 S
888 889 0 3 ... 23.45 NaN S	888 889 0 3 ... 23.45 NaN S
889 890 1 1 ... 30.00 C148 C	889 890 1 1 ... 30.00 C148 C
890 891 0 3 ... 7.75 NaN Q	890 891 0 3 ... 7.75 NaN Q

```
[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
       PassengerId   Survived      Pclass  ...       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  ...  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642  ...    0.523008    0.381594   32.204208
std     257.353842    0.486592    0.836071  ...    1.102743    0.806057   49.693429
min       1.000000    0.000000    1.000000  ...    0.000000    0.000000    0.000000
25%     223.500000    0.000000    2.000000  ...    0.000000    0.000000    7.910400
50%     446.000000    0.000000    3.000000  ...    0.000000    0.000000   14.454200
75%     668.500000    1.000000    3.000000  ...    1.000000    0.000000   31.000000
max     891.000000    1.000000    3.000000  ...    8.000000    6.000000  512.329200

[8 rows x 7 columns]
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
```


```
[5 rows x 12 columns]
(891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 66.2+ KB
None
       PassengerId   Survived      Pclass  ...       SibSp       Parch        Fare
count   891.000000  891.000000  891.000000  ...  891.000000  891.000000  891.000000
mean    446.000000    0.383838    2.308642  ...    0.523008    0.381594   32.204208
std     257.353842    0.486592    0.836071  ...    1.102743    0.806057   49.693429
min       1.000000    0.000000    1.000000  ...    0.000000    0.000000    0.000000
25%     223.500000    0.000000    2.000000  ...    0.000000    0.000000    7.910400
50%     446.000000    0.000000    3.000000  ...    0.000000    0.000000   14.454200
75%     668.500000    1.000000    3.000000  ...    1.000000    0.000000   31.000000
max     891.000000    1.000000    3.000000  ...    8.000000    6.000000  512.329200

[8 rows x 7 columns]
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
```

4.2.6. Titanic Dataset Analysis and Data Cleaning - 2

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Create a new column ‘IsAlone’ which is 1 if the passenger is alone (FamilySize = 0), otherwise 0.

Convert the ‘Sex' column to numeric values (male: 0, female: 1).

One-hot encode the ‘Embarked’ column, dropping the first category.

Get the mean age of passengers.

Get the median fare of passengers.

Get the number of passengers by class.

Get the number of passengers by gender.

Get the number of passengers by survival status.

Calculate the survival rate of passengers.

Calculate the survival rate by gender.

The Titanic dataset contains columns as shown below :

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
```

```
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd import numpy as np  # Load the Titanic

dataset data = pd.read_csv('Titanic-Dataset.csv')

data['FamilySize'] = data['SibSp'] + data['Parch']

 # 1. Create a new column ‘IsAlone' (1 if alone, 0 otherwise) data['IsAlone'] =

np.where(data['FamilySize']==0,1,0)  # 2. Convert ‘Sex' to numeric (male: 0, female: 1)

data['Sex']=data['Sex'].map({'male':0, 'female':1}) # 3. One-hot encode the ‘Embarked'

column data =pd.get_dummies(data, columns = ['Embarked'], drop_first=True)

 # 4. Get the mean age of passengers

mean_age=data['Age'].mean() print(mean_age)

 # 5. Get the median fare of passengers m

=data['Fare'].median() print(m)

 # 6. Get the number of passengers by class  p

=data['Pclass'].value_counts() print(p)

 # 7. Get the number of passengers by gender k =

data['Sex'].value_counts() print(k)

 # 8. Get the number of passengers by survival status s =

data['Survived'].value_counts() print(s)  sk=data['Survived'].mean()

print(sk)  sg = data.groupby('Sex')['Survived'].mean() print(sg)
```

Output:

Expected output	Actual output
29.69911764705882	29.69911764705882
14.4542	14.4542
3 491	3 491
1 216	1 216
2 184	2 184
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
0 577	0 577
1 314	1 314
Name: Sex, dtype: int64	Name: Sex, dtype: int64
0 549	0 549
1 342	1 342
Name: Survived, dtype: int64	Name: Survived, dtype: int64
0.3838383838383838	0.3838383838383838
Sex	Sex
0 0.188908	0 0.188908
1 0.742038	1 0.742038
Name: Survived, dtype: float64	Name: Survived, dtype: float64

4.2.7. Titanic Dataset Analysis and Data Cleaning - 3

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Calculate the survival rate by class.

Calculate the survival rate by embarkation location (Embarked_S).

Calculate the survival rate by family size (FamilySize).

Calculate the survival rate by being alone (IsAlone).

Get the average fare by passenger class (Pclass).

Get the average age by passenger class (Pclass).

Get the average age by survival status (Survived).

Get the average fare by survival status (Survived).

Get the number of survivors by class (Pclass).

Get the number of non-survivors by class (Pclass).

The Titanic dataset contains columns as shown below

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

```
import pandas as pd import numpy as np  # Load the Titanic dataset data =
pd.read_csv('Titanic-Dataset.csv') data['FamilySize'] = data['SibSp'] + data['Parch']
data['IsAlone'] = np.where(data['FamilySize'] > 0, 0, 1) data = pd.get_dummies(data,
columns=['Embarked'], drop_first=True)
 # 1. Calculate the survival rate by class
print(data.groupby('Pclass')['Survived'].mean())  # 2. Calculate the
survival rate by embarked location
print(data.groupby('Embarked_S')['Survived'].mean())  # 3.
Calculate the survival rate by family size
```

```
print(data.groupby('FamilySize')['Survived'].mean())  # 4. Calculate
the survival rate by being alone
print(data.groupby('IsAlone')['Survived'].mean())
 # 5. Get the average fare by class
print(data.groupby('Pclass')['Fare'].mean())  # 6. Get the
average age by class
print(data.groupby('Pclass')['Age'].mean()) # 7. Get the
average age by survival status
print(data.groupby("Survived")['Age'].mean())  # 8. Get
the average fare by survival status
print(data.groupby('Survived')['Fare'].mean())  # 9. Get
the number of survivors by class
sbc=data[data['Survived']==1]['Pclass'].value_counts()
print(sbc)
 # 10. Get the number of non-survivors by class
nbc=data[data['Survived']==0]['Pclass'].value_counts() print(nbc)
```

Output:

Test case 1 163 ms Debug

Expected output	Actual output
Pclass	Pclass
1 0.629630	1 0.629630
2 0.472826	2 0.472826
3 0.242363	3 0.242363
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Embarked_S	Embarked_S
0 0.506073	0 0.506073
1 0.336957	1 0.336957
Name: Survived, dtype: float64	Name: Survived, dtype: float64
FamilySize	FamilySize
0 0.303538	0 0.303538
1 0.552795	1 0.552795
2 0.578431	2 0.578431
3 0.724138	3 0.724138
4 0.200000	4 0.200000
5 0.136364	5 0.136364

6 0.333333	6 0.333333
7 0.000000	7 0.000000
10 0.000000	10 0.000000
Name: Survived, dtype: float64	Name: Survived, dtype: float64
IsAlone	IsAlone
0 0.505650	0 0.505650
1 0.303538	1 0.303538
Name: Survived, dtype: float64	Name: Survived, dtype: float64
Pclass	Pclass
1 84.154687	1 84.154687
2 20.662183	2 20.662183
3 13.675550	3 13.675550
Name: Fare, dtype: float64	Name: Fare, dtype: float64
Pclass	Pclass
1 38.233441	1 38.233441
2 29.877630	2 29.877630
3 25.140620	3 25.140620
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 30.626179	0 30.626179
1 28.343690	1 28.343690
Name: Age, dtype: float64	Name: Age, dtype: float64
Survived	Survived
0 22.117887	0 22.117887
1 48.395408	1 48.395408
Name: Fare, dtype: float64	Name: Fare, dtype: float64
1 136	1 136
3 119	3 119
2 87	2 87
Name: Pclass, dtype: int64	Name: Pclass, dtype: int64
3 372	3 372
2 97	2 97
1 80	1 80

4.2.8. Titanic Dataset Analysis and Data Cleaning - 4

You are provided with the Titanic dataset containing information about passengers on the Titanic. Your task is to write Python code to answer the following questions based on the dataset.

Get the number of survivors by gender (Sex).

Get the number of non-survivors by gender (Sex).

Get the number of survivors by embarkation location (Embarked_S).

Get the number of non-survivors by embarkation location (Embarked_S).

Calculate the percentage of children (Age < 18) who survived.

Calculate the percentage of adults (Age >= 18) who survived.

Get the median age of survivors.

Get the median age of non-survivors.

Get the median fare of survivors.

Get the median fare of non-survivors.

The Titanic dataset contains columns as shown below :

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked

Sample Data:

```
PassengerId,Survived,Pclass,Name,Sex,Age,SibSp,Parch,Ticket,Fare,Cabin,Embarked
1,0,3,"Braund, Mr. Owen Harris",male,22,1,0,A/5 21171,7.25,,S
2,1,1,"Cumings, Mrs. John Bradley (Florence Briggs Thayer)",female,38,1,0,PC
17599,71.2833,C85,C
3,1,3,"Heikkinen, Miss. Laina",female,26,0,0,STON/O2. 3101282,7.925,,S
4,1,1,"Futrelle, Mrs. Jacques Heath (Lily May Peel)",female,35,1,0,113803,53.1,C123,S
5,0,3,"Allen, Mr. William Henry",male,35,0,0,373450,8.05,,S
6,0,3,"Moran, Mr. James",male,,0,0,330877,8.4583,,Q
7,0,1,"McCarthy, Mr. Timothy J",male,54,0,0,17463,51.8625,E46,S
8,0,3,"Palsson, Master. Gosta Leonard",male,2,3,1,349909,21.075,,S
9,1,3,"Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)",female,27,0,2,347742,11.1333,,S
10,1,2,"Nasser, Mrs. Nicholas (Adele Achem)",female,14,1,0,237736,30.0708,,C
```

Code:

import pandas as pd import

numpy as np

Load the Titanic dataset

data = pd.read_csv('Titanic-Dataset.csv') data = pd.get_dummies(data,

columns=['Embarked'], drop_first=True)

1. Get the number of survivors by gender

sbg=data[data['Survived']==1]['Sex'].value_counts() print(sbg)

2. Get the number of non-survivors by gender

nsbg=data[data['Survived']==0]['Sex'].value_counts() print(nsbg)

3. Get the number of survivors by embarked location

nse=data[data['Survived']==1]['Embarked_S'].value_counts() print(nse)

4. Get the number of non-survivors by embarked location

a=data[data['Survived']==0]['Embarked_S'].value_counts() print(a)

5. Calculate the percentage of children (Age < 18) who survived

c=data[data['Age']<18] print(c['Survived'].mean())

6. Calculate the percentage of adults (Age >= 18) who survived

a=data[data['Age']>=18] print(a['Survived'].mean())

7. Get the median age of survivors print(data[data['Survived']==1]['Age'].median())

8. Get the median age of non-survivors print(data[data['Survived']==0]['Age'].median()) # 9. Get the median fare of survivors print(data[data['Survived']==1]['Fare'].median()) # 10. Get the median fare of non-survivors print(data[data['Survived']==0]['Fare'].median())

Output:

Expected output

```
female    233
male      109
Name: Sex, dtype: int64
male      468
female     81
Name: Sex, dtype: int64
1    217
0    125
Name: Embarked_S, dtype: int64
1    427
0    122
Name: Embarked_S, dtype: int64
0.5398230088495575
0.3810316139767055
28.0
28.0
26.0
```

Actual output

```
female    233
male      109
Name: Sex, dtype: int64
male      468
female     81
Name: Sex, dtype: int64
1    217
0    125
Name: Embarked_S, dtype: int64
1    427
0    122
Name: Embarked_S, dtype: int64
0.5398230088495575
0.3810316139767055
28.0
28.0
26.0
```